Senior Health Workforce Analyst - Assignment

Health Workforce Modelling

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About me

- Applied quantitative modeler with 10+ years of experience:
  - Decision models (optimization)
  - Agent-based simulation
  - Physics-based modelling
  - Macroeconomic models
- Currently, Data Analytics Specialist @ MCCSS.
- Goals:
  - Apply data analytics and modelling to inform decision-making and uncover actionable insights.
  - Learn and grow as a professional by taking on new challenges and responsibilities.

Assignment & Modelling Approach

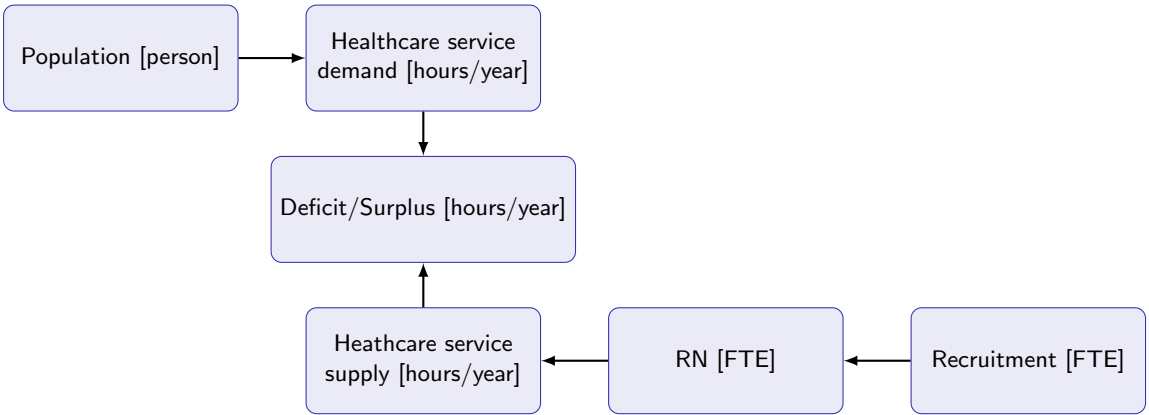
Develop a quantitative model to forecast workforce supply and workforce requirements for **Registered Nurses** (RNs) in Ontario over the next 10 years.

Modelling approach:

- Start simple and enrich only if needed.
- Research or policy question being addressed (how a solution looks like?).
- Decision variables or policy instruments (what can be controlled?).
- Causal relationships between variables.

There is no good solution to a badly posed problem.

Conceptual formulation



- FTE: Full-time equivalent.

Model Decisions

- Which variables are considered exogenous?
- Spatial and temporal granularity.
- Most appropriate model type:
  - System dynamics
  - Time series
  - Optimization
  - Agent-based modelling
- Model should drive data sources (rather than the other way around).
- Look for solutions in literature.
- Start with a simple model first.

Demand-side Modelling

- Demand of healthcare services: d_t [hours/time]
- α: per capita demand for healthcare services [hours/(person time)]
- Population-driven demand:

$$d_t = \alpha \; p_t \quad \forall t \qquad \text{[hours/time]}$$

- Age-based demand: $d_t = \sum_a \alpha_a \; p_{a,t}$
- Specialty-based demand: $d_{t,s} = \sum_a \alpha_{a,s} \; p_{a,t}$

Population Modelling

- Leslie (age-structured) model, in vector form:

$$\mathbf{p}_{t+1} = \mathbf{L}\ \mathbf{p}_t + \mathbf{i}_t \quad \forall t$$

- $\mathbf{p}_t$: population vector at time t.
- $\mathbf{L}$: Leslie matrix.
- $\mathbf{i}_t$: net immigration vector at time t (major source of uncertainty).

Supply-side Modelling

- Supply of healthcare services: s_t [hours/time]
- β_t: per capita supply of healthcare services [hours/(FTE time)]
- Supply of healthcare services:

$$s_t = \beta \; n_t \quad \forall t \qquad \text{[hours/time]}$$

- Specialty-based supply: $s_{t,s} = \beta_s \; n_{s,t}$

Balance Surplus/Deficit

- Balance of healthcare services:

$$y_t = d_t - s_t \quad \forall t \qquad \text{[hours/time]}$$

- Surplus: $y_t^+ = \max(y_t, 0)$
- Deficit: $y_t^- = \max(-y_t, 0)$

Optimal hiring policy model

Decision variable x_t: new RNs hired at time t.

$$\min_{x_t} \sum_{t=0}^{T} \lambda \; e^{-\delta t} \, y_t^+ + \sum_{t=0}^{T} \mu \; e^{-\delta t} \, y_t^-$$

s.t.

$$d_t = \alpha_t \; p_t \quad \forall t \tag{1}$$

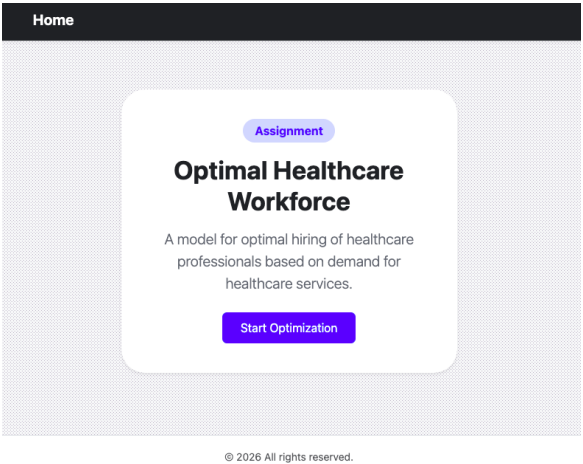
$$s_t = \beta_t \; n_t \quad \forall t \tag{2}$$

$$d_t - s_t = y_t^+ - y_t^- \quad \forall t \tag{3}$$

$$n_t = \gamma \; n_{t-1} + x_t \quad \forall t \tag{4}$$

$$x_t, y_t^+, y_t^- \geq 0 \quad \forall t \tag{5}$$

Decision Support Tool



Decision Support Tool Model: https://20.220.16.0/

Development tools

- Model formulation: OR-Tools
- Application framework: Flask
- Infrastructure:
  - Azure virtual machine (Ubuntu 24.04)
  - Nginx, Gunicorn
- Available at `https://20.220.16.0/`

Data sources, variables and indicators

	Sources	**Variables / indicators**
Supply	College of Nurses of Ontario (CNO) registration statistics; CIHI Nursing in Canada / National Nursing Database; program enrolment and graduation counts; licensing exam results	Headcount and FTE; age and sex; region; practice setting; employment status (full-/part-time); graduation, pass and licensing rates; migration
Requirements	Ontario Ministry of Finance population projections; CIHI health service utilization; Ontario Health regional reports; Statistics Canada	Population by region/age/sex; service use by setting; RNs per 1,000 population; hours of care per service unit
Policy	Training-seat funding, IEN pathway capacity, program costs	Unit cost per seat, IEN licence, retention incentive; capacity limits; ramp-up rates

Data quality, gaps and privacy

Quality and gaps

- Registration overstates active, practising supply → adjust using employment status and FTE conversion
- Small regional cells are suppressed or unstable → pool years or regions
- 1–2 year publication lag → nowcast the latest year and flag it
- Self-reported hours are noisy → validate against multiple sources
- Reconcile definitions across CNO, CIHI and StatCan; document every assumption

Privacy

- Aggregate, de-identified data only
- No individual licensing or patient records needed
- Respect CIHI/CNO data-use terms and small-cell suppression rules
- Follow FIPPA principles for any internal administrative data

Assessing uncertainty: scenarios and sensitivity

Scenarios

- **Baseline:** status-quo trends
- **High attrition:** accelerated retirements
- **Expanded seats:** policy intervention
- **High IEN:** fast-tracked licensing / immigration

Sensitivity analysis

- One-at-a-time variation of attrition, utilization growth, completion/pass rates and migration $\rightarrow$ tornado chart of which assumptions move the 2036 gap most
- Monte Carlo draws from plausible parameter ranges $\rightarrow$ uncertainty band around supply and requirements

Robust recommendations

Re-solve the optimization under each scenario: show how the policy mix and its cost change, and favour options that perform well across scenarios.

Communicating results to senior management

1. **Headline chart:** projected supply vs. requirements, 2026–2036, province-wide and by region, with a shaded uncertainty band.
2. **Regional gap view:** when and where gaps first appear.
3. **Optimization result:** the lowest-cost policy mix (seats, IEN, retention, by year) as a stacked bar chart.
4. **Tornado chart:** which assumptions matter most, and so where better data is worth the effort.

- Plain language, one message per slide, assumptions stated up front.
- Brief for executives; technical appendix for analysts; reproducible code.

Limitations and policy implications

Limitations

- Quality depends on data lag and assumption validity
- Does not capture non-linear shocks (e.g. pandemics)
- Regional model is coarser than facility-level planning
- Requirements reflect current care models unless scenarios change them

Policy options (with trade-offs)

- Expand training seats: durable but slow
- Fast-track IEN licensing: quick, limited by pathway capacity
- Retention incentives: fast, recurring cost
- Regional / rural incentives: target maldistribution
- Scope-of-practice / skill-mix change: shifts requirements

Each option is presented with cost, lead time and feasibility to inform government action.

Summary

- A linked **supply** and **needs-based requirements** model, by region, over ten years.
- An **optimization layer** turns gaps into a least-cost policy mix.
- **Scenarios and sensitivity analysis** make recommendations robust.
- Built on public, aggregate data, with explicit quality and privacy safeguards.
- Communicated as options with trade-offs, refreshed annually as new data arrive.

Thank you — questions?